
The Instrument Is the First Suspect: Six Broken Lenses in One Empirical Program

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Abstract

1 Empirical claims about learned systems inherit the health and calibration of the
2 instruments that measure them. We report six pre-publication incidents from one
3 measurement-heavy research program on fast-weight sequence models in which
4 a trained model appeared to fail and the failure traced instead to the instrument:
5 an admission tolerance calibrated on the wrong key population; a probe reading
6 a layer causal state-zeroing later proved inert; an identity-gauge assumption mea-
7 sured on one architecture and silently carried to another; an uncentered covariance
8 estimator degenerate on the task’s own near-orthogonal targets; a target whose am-
9 bient identity block taxed every rank budget; and a cross-check reference invoked
10 on the transposed side of a state convention. Each incident was caught by one
11 and the same discipline: instrument-health adjudication before any model verdict,
12 pre-registered crosschecks graded where the primary lens’s assumptions fail, fal-
13 sifier controls run to completion, and raw-artifact tiebreaks. The discipline is itself
14 falsifiable; its decisive crosscheck survived a pre-registered shuffled-target control
15 that would have voided the adjudication. It does not launder results: the corrected
16 lens left the pre-registered endpoint verdict standing and two genuine model fail-
17 ures intact. We argue the discipline transfers to any empirical evaluation gated by
18 numerical checks, learned probes, or basis-dependent readouts.

19 1 Introduction

20 A measurement-heavy empirical program breaks its instruments more often than its models. Within
21 five days, one research program on fast-weight sequence models produced six apparent headline
22 model failures: a capacity collapse (§3); three consecutive failed evaluation rounds (§4); a falsified
23 generalization endpoint (§5); a perfect synthetic model rejected by its acceptance test; a causal rank
24 test reading zero at every rank; and a recurrence contradicted by a reference kernel (§6). In every
25 one, a fixed adjudication procedure traced the failure to the instrument, not the model (Table 1);
26 in two further readings the same procedure confirmed the model, not the lens, had failed. The
27 asymmetry is the point: an evaluation pipeline that never suspects its own instruments will publish
28 lens artifacts as model results.

29 Our contributions: (1) the catalogue, each incident with its mechanism and a reusable diagnostic
30 signature; (2) an explicit five-rule adjudication discipline (§2) that caught all six before publication;
31 (3) the discipline’s boundary: it did not flip the pre-registered endpoint verdict, and two genuine
32 model failures survived every instrument fix (§7).

33 2 The Adjudication Discipline

34 None of the five rules below is novel in isolation; applied together, *before* any model verdict is
35 recorded, they form a lens-failure detector with a measured record.

36 **Rule 1: instrument-health adjudication precedes model verdicts.** Before a sweep’s pre-
 37 registered endpoint may be read, the harvest walks a health checklist over the raw per-cell artifacts:
 38 gate violations, convergence, and a contradiction scan between independent readouts (the rule that
 39 fired in §5).

40 **Rule 2: pre-register a crosscheck that removes the primary lens’s strongest assumption,** graded
 41 wherever that assumption provably fails, with the switching rule written down in advance.

42 **Rule 3: falsifier teeth, run to completion.** Every negative control is executed, not merely writ-
 43 ten; guards are mutation-tested; every probe rung carries a planted-signal positive control; and the
 44 adjudication itself carries a pre-registered falsifier whose firing voids it.

45 **Rule 4: contradictions are resolved against raw artifacts.** When two reads disagree, neither
 46 recency nor authority decides; the adjudicator recomputes both from the committed raw files and
 47 records the tiebreak before anything downstream runs.

48 **Rule 5: argmax closure.** Wherever a claim requires exact continuous recovery, the grading path
 49 must not be satisfiable by argmax decoding: under argmax a rank-1 matrix can recover on the order
 50 of d associations [Nichani et al., 2025]. The converse gap was pre-registered before it fired (§4).

51 3 Case I: The Tolerance That Manufactured a Cliff

52 A fast-weight associative memory’s recall capacity is measured by exact continuous recovery (cosine
 53 ≥ 0.9 , never argmax; Rule 5) across a load sweep K/d_{state} ; an eval-time admission gate decides
 54 which trained cells may enter a capacity fit, one leg checking that a Newton–Schulz orthogonaliza-
 55 tion inside the scoring path converges to residual < 0.01 within $n_{\text{iter}} = 20$ iterations. A wide-grid
 56 sweep at $d_{\text{state}} = 96$ trained cleanly at every cell (the training guard read exactly 1.0 throughout),
 57 yet of 12 new cells only one was admissible, with $K=78, 84$, and 90 failing on all three seeds each:
 58 the shape of a capacity cliff, contradicting the same architecture’s established super-linear growth at
 59 lower state sizes [Anonymous, 2026].

60 The first mechanistic diagnosis blamed the trained anchor table’s convergence at high load; a follow-
 61 up refuted it with margins (residuals near 1.2×10^{-6} , roughly $7,800\times$ below tolerance, identical
 62 between failing and passing cells, failures confined to one pool the anchor table architecturally
 63 bypasses) and the claim was formally retracted.

64 A purpose-built repro instrument then recovered the true failing tensor: the pool’s raw keys, never
 65 saved by the original checkpoint writer. Its decision rule was pre-registered as a strict precedence
 66 order and walked mechanically (Table 2); every instrument-bug branch failed to fire, and an iteration
 67 sweep resolved every one of the 4,608 flagged episodes by $n_{\text{iter}} \leq 28$. A wall keeps failing as the
 68 budget grows; a miscalibrated tolerance resolves at a fixed bump, and did. Root cause: a tolerance
 69 calibrated against anchor-stabilized *blended* keys, silently applied to the structurally different *raw*
 70 key population this one pool queries. Only the admission flag was recalibrated, only for the 11
 71 cells verified to share the identical signature, at zero additional GPU cost; the re-fit found no cliff
 72 through $K/d_{\text{state}} = 0.9375$ (per-load mean recovery 0.92 to 1.00).

73 4 Case II: The Instrument at the Wrong Layer

74 A two-block fast-weight model is trained on an associative binding task ($K = 32$ bindings per
 75 episode) against matched baselines, with two registered readouts: a discrete task metric (episode-
 76 restricted recall through the model’s own continuation and LM head) and a continuous probe metric
 77 $\text{rf}@0.9$ (the fraction of targets a linear probe on a tapped state recovers at cosine ≥ 0.9). In the third
 78 of three consecutive rounds recorded as failures at $\text{rf}@0.9 = 0.0$, the discrete metric dissociated:
 79 recall accuracy 0.9990 against chance 0.03125 (baselines 0.0447 and 0.0295), the probe still exactly
 80 0.0 in every arm. The dissociation, not the zero, is the finding (Rule 5).

81 An offline re-fit battery removed the probe-training explanations: closed-form ridge regression, the
 82 global optimum for the linear probe class, reproduced $\text{rf}@0.9 = 0.0$ exactly, as did pinned SGD and
 83 MLP probes; the trained probe’s output was the analytically known episode-membership direction
 84 (mean cosine 0.176 against the $1/\sqrt{32} \approx 0.177$ ceiling; membership cosine 0.896); the ridge fit
 85 cleared its shuffled-tap control by at most +0.059 cosine. Meanwhile the full forward pass decodes

the answer at 0.9957 accuracy ($31.9\times$ chance) for the contender, both baselines at or near chance: the information exists in the network and is not linearly present at the tapped state.

A state-zeroing experiment located the storage (Figure 2): zeroing the first block’s state S_0 collapses recall to chance (0.0286); zeroing S_1 , the layer every deployed instrument tapped, leaves it bit-unchanged at 0.9990: the probed layer is causally inert for the probed behavior. Placement alone is not the whole defect: a ridge probe on the load-bearing S_0 *also* reads $\text{rf}@0.9 = 0.0$ (gaps over shuffled controls $+0.002$ to $+0.063$ across all state-level taps), and only the post-block-1, pre-LM-head hidden exposes the recall linearly ($\text{rf}@0.9 = 0.674$, gap $+0.800$), only in the arm that can perform the task (the ablation’s same tap reads 0.0; Table 3). The bindings are stored in a format no tested linear read on either state exposes; the model’s own downstream nonlinearity is the only decoder found.

The same diagnosis exposed a second defect in passing: an auxiliary classifier whose labels were, by construction, uniform given the decoded quantity, so a perfect tap scores chance; its expected-null had been read as confirmation for a round, hence Rule 3’s positive-control clause.

5 Case III: The Gauge That Did Not Transfer

A recurrent composer is trained on the word problems of five finite groups and scored for compositional depth generalization: trained at depth ≤ 8 , tested at depth 64. Two pre-registered lenses score recovery of the group representation from the state: the *primary* de-gauges with a scale-only correction, valid when the learned basis matches the target basis up to scale ($\hat{Q} \approx I$), measured to hold on every checkpoint of the *previous* architecture; the *crosscheck* fits a full orthogonal intertwiner by Procrustes on a fit split and scores on a disjoint eval split, assuming nothing about the gauge.

The signature. The 62-cell sweep completed cleanly, and the endpoint machinery, run as built on the primary lens, returned FALSIFY: no measurable separation from the baseline at the decisive groups. The mandatory instrument-health pass (Rule 1) then surfaced a contradiction (Figure 1): wherever training had *not* converged, both lenses read near zero; wherever it converged, they contradicted each other at full magnitude (one converged cell: final loss 0.0001, primary recovery 0.050, crosscheck 1.000). All 13 cells with a lens disagreement of at least 0.5 are converged cells; on all 30 baseline cells the lenses agree exactly. The harvest stopped for adjudication with no model verdict recorded.

The tiebreak. A leaky crosscheck reading high on junk would leave the falsification standing, so either lens could be the broken one. The adjudication, performed against raw artifacts and code (Rule 4), rested on four grounds: (1) the crosscheck is leakage-guarded by construction (fit on the fit split, scored on the disjoint eval split); (2) it demonstrably discriminates, reading near zero on every non-converged cell; (3) the primary’s scale-only correction was already flagged as basis-brittle across seeds (per-seed mean cosine 0.03 to 0.69 where the crosscheck read 0.86 to 0.95), and its $\hat{Q} \approx I$ evidence came from a different architecture; a perfect model in a rotated basis reads zero under an identity gauge by construction; (4) the synthetic acceptance test already grades the crosscheck when the injected gauge is non-identity (Rule 2). Verdict: the primary lens is broken on converged cells; the mechanical falsification is void *as a model verdict*.

Teeth. The tiebreak carried its own pre-registered falsifier (Rule 3): re-scoring with shuffled targets must read near zero, or the tiebreak is wrong and the contradiction escalates to an instrument rebuild. On three converged checkpoints the shuffled crosscheck read 0.00, 0.00, and 0.05, far below the 0.5 void threshold, while the unshuffled reads reproduced the committed values bit-identically (Table 5).

The re-metric. The full grid was re-read under the crosscheck lens through the unmodified verdict machinery, which first reproduced the committed primary verdict and all 62 recomputed ceilings to within 10^{-9} . The corrected endpoint is *still* FALSIFY, for a more informative reason (§7, Table 4): not “no signal anywhere” but “two specific, lens-independent failures at the two decisive groups.” Fixing the instrument changed the reason, not the verdict.

6 Three More Lenses, Briefly

Three further incidents, detailed in Appendix A, round out the catalogue: an **uncentered covariance**, degenerate on the task’s own near-orthogonal targets ($ZZ^\top \approx c^2 I$), under which a synthetic *perfect* model failed the production acceptance bars (mean cosine 0.084 versus 0.9996 centered); a **target with an ambient identity block** that taxed every rank budget, so a causal rank experiment

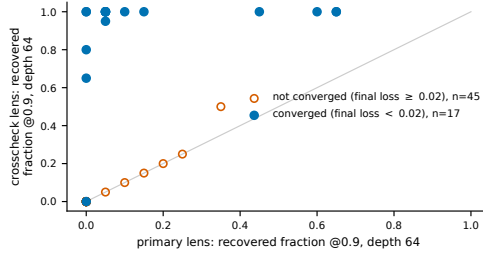


Figure 1: Primary versus crosscheck recovery (fraction of items recovered at cosine ≥ 0.9 , depth 64), all 62 sweep cells. Non-converged cells (open orange) sit near the origin under both lenses; every cell with a lens disagreement of at least 0.5 is a converged cell (filled blue). A lens defect that activates exactly where the model succeeds is the inverse of noise.

137 returned zeros at every rank while the capped arms sat at their exact theoretical ceiling $\sqrt{k/d_{\text{state}}}$
 138 (37 of 39 cells within 0.07): undelivered by the instrument, not failed by the model; and a **reference**
 139 **kernel invoked on the transposed side** of a state convention, “contradicting” a correct recurrence
 140 at relative residual $1.40 \approx \sqrt{2}$, the exact signature of a transposed rank-1 outer product, exonerated
 141 by a closed-form hand computation at 4.5×10^{-8} .

142 7 The Boundary: What the Discipline Did Not Flip

143 A lens-repair procedure that only ever rescues the researcher’s preferred hypothesis is a laundering
 144 machine. Under the corrected lens, Case III’s endpoint remained **FALSIFY**, held there by two lens-
 145 independent model results: one decisive configuration never reaches training-support convergence
 146 (recovery 0.00 at depth 64 for all five seeds, both lenses agreeing), and the other decisive group sits
 147 below its pre-registered bar (0.483 against 0.735) because one seed trains to low loss yet generalizes
 148 at 0.00 under both lenses while its siblings individually clear 80% and 65% of their own ceilings
 149 against a baseline pinned at exactly 0.0.

150 Case I’s follow-ups show the same refusal: a diagnosed pool-margin confound moved the measure-
 151 ment by $+0.0330$, roughly $12\times$ its measured noise floor of 0.0028, yet was withheld because the
 152 diagnostic’s own cells never cleared admission; and an archived exact-ceiling reading (1.0000 on all
 153 three original seeds) did not replicate at a fresh seed (0.9725) and was re-scoped to the seeds mea-
 154 sured. The machinery that unlocked eleven quarantined cells declined, twice, to promote findings it
 155 could not support.

156 8 Related Work and Limitations

157 **Related work.** Silberzahn et al. [2018] and D’Amour et al. [2022] document divergent conclusions
 158 from underspecified analysis pipelines; here the models are fixed and the *measurement pipeline*
 159 is the component being adjudicated, with a per-incident decision procedure rather than a variance
 160 demonstration. Henderson et al. [2018] and Bouthillier et al. [2021] quantify seed and tuning
 161 variance; none of the six defects here is sampling variance (a transposed convention or a wrong-
 162 population tolerance is bit-stable across seeds), so variance accounting would have caught none.
 163 REFORMS [Kapoor et al., 2023], Sculley et al. [2018], and Lipton and Steinhardt [2018] prescribe
 164 what a trustworthy paper discloses; this paper supplies the in-flight machinery such checklists
 165 presume exists. Nichani et al. [2025] prove the argmax/exact-recovery gap of Rule 5; we report it
 166 firing live, on both sides (§4); the fast-weight testbeds follow Schlag et al. [2021].

167 **Limitations.** All six incidents come from one program, one team, one architecture family; we
 168 cannot report the rate a lighter process would miss, and six incidents support no prevalence claim.
 169 The catalogue is survivor-biased in one stated direction: a defect that both the crosscheck and the
 170 falsifier battery miss would appear in no record we could write. The economics still favor adoption:
 171 Case I’s two decisive diagnostic instruments together cost under 2 GPU-hours against the 6.33
 172 GPU-hours of training their uncorrected readings would have discredited; auditing the lens is cheap
 173 relative to the compute, and the claims, it protects.

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A The Three Brief Lenses, in Full

The covariance that could not see. An acceptance test for a representation-recovery pipeline injected a synthetic *perfect* model and failed it. The subspace step estimated an uncentered covariance, degenerate on the task’s own targets: for near-orthogonal states $Z \approx c(\rho \oplus I)$, $ZZ^\top \approx c^2I$ is isotropic and carries no subspace information. On the production harness, an oracle injection scored mean cosine 0.084 and recovery 0.05 under the uncentered lens versus 0.9996 and 1.00 once the covariance was centered; the same artifact records the scale-only primary reading 0.046 on data whose true gauge is non-identity while the full-intertwiner crosscheck reads 0.9996, the pre-echo of §5. The fix (centering) cancels the constant identity block because the group mean of a nontrivial representation vanishes; the acceptance injection was extended so the previously skipped subspace step now sits inside the test.

The target that taxed every budget. A causal test of a rank-recruitment law capped model rank at k around the predicted minimum $d_{\min}$ and expected recovery to return at $k = d_{\min}$. Every capped arm returned zero, at every group, on both lenses. The target had been constructed as $\rho \oplus I$: the ambient identity block inflates the as-built target rank by two, so every rank budget buys the constant block first. The diagnosis is quantitative: a rank- k model’s best achievable cosine against this target is exactly $\sqrt{k/d_{\text{state}}}$, and 37 of 39 force-rank cells sat within 0.07 of that ceiling (mean absolute deviation 0.028); the arms had trained to their theoretical optimum against the wrong target. The causal test was undelivered by the instrument, not failed by the model; a zero-padded re-run subsequently delivered it [Anonymous, 2026].

The reference on the wrong side. A bespoke recurrence was cross-checked against a reference kernel and disagreed at relative Frobenius residual 1.40; the recurrence was the initial suspect. At depth one the update is closed-form ($S_1 = \beta vk^\top$), so a hand computation with scalar loops adjudicates conventions with no numerics: the composer matched the closed form at 4.5×10^{-8} and its *transpose* at 1.405, and for near-orthogonal unit k the transpose signature is exactly $\|vk^\top - kv^\top\|_F / \|vk^\top\|_F \approx \sqrt{2}$. The reference library’s documented state convention is the transpose of the pinned equation; the cross-check’s own reference invocation was the defect. With the returned state transposed before comparison, the cross-check passes all three pinned configurations (residuals 2.8×10^{-3} to 4.5×10^{-3}), and a permanent negative test confirms a transposed-update mutant is killed at 1.405.

B The Catalogue and Per-Incident Detail Tables

Table 1 is the six-incident catalogue in one view; the remaining tables and Figure 2 carry the per-incident numbers the body summarizes. Every value traces to the same evidence rows (and raw artifacts, with checksums) recorded in the paper’s claims-to-evidence map.

Apparent model failure	Actual instrument defect	Diagnostic signature	Resolution
Capacity collapse, 11/12 cells quarantined (§3)	admission tolerance calibrated on blended keys, applied to raw keys	every flagged episode resolves under a modest iteration bump	flag-only recalibration; no cliff
Three failed evaluation rounds (§4)	probe tap on a causally inert layer; storage not linearly decodable at any state	task metric near ceiling while probe reads zero; state-zeroing localization	metric of record re-registered; tap relocated
Generalization endpoint falsified with no signal anywhere (§5)	identity-gauge assumption from one architecture carried to another	primary/crosscheck contradiction of full magnitude, only on converged cells	crosscheck decisional; verdict re-read (still falsified, different reason)
Perfect synthetic model fails acceptance (§6)	uncentered covariance, degenerate on near-orthogonal targets	oracle injection scores near zero; centering restores it	estimator centered; injection extended
Causal rank test returns zeros at every rank (§6)	target built with an ambient identity block, taxing every rank budget	capped cells sit at the exact ceiling $\sqrt{k/d_{\text{state}}}$	target zero-padded; test re-run
Recurrence contradicted by reference kernel (§6)	reference invoked on the transposed side of a state convention	relative residual $\approx \sqrt{2}$; closed-form hand computation	reference transposed; mutant-kill check added

Table 1: The six incidents. In each, the model was initially the suspect and the instrument was the finding; every row was adjudicated against raw artifacts before any model-level conclusion was recorded. Sections named per row.

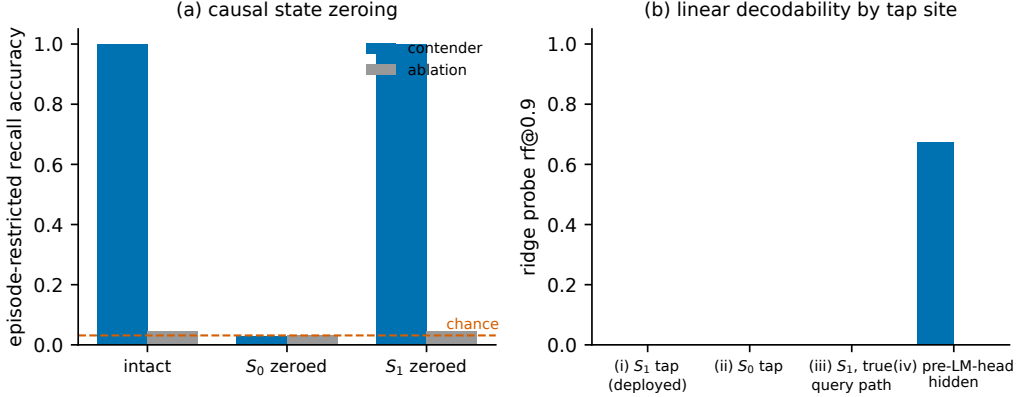


Figure 2: Case II, the wrong-layer instrument adjudicated causally. (a) Episode-restricted recall accuracy under state zeroing, contender (blue) and ablation baseline (gray), chance = $1/32$ dashed: zeroing S_0 collapses recall to chance in both arms; zeroing S_1 , the layer every deployed probe tapped, changes nothing. (b) Linear (ridge) probe recovery rf@0.9 by tap site: both recurrent states read zero in both arms, including the causally load-bearing S_0 ; only the post-nonlinearity pre-LM-head hidden exposes the recall, and only in the arm that can perform the task. The figure regenerates from the archived localization summary under an asserted checksum.

Step (pre-registered order)	Outcome	Reading	Rules out
entity-set reconstruction gate	pass	107/107 set-equal	wrong-partition analysis
pool-membership precheck	pass	0 violations / 36 events	instrument bug (structural)
reproduction-count floor	pass	$12\times$ the minimum	flake event
live/offline agreement	pass	0 disagreements	instrument bug (numerical)
iteration sweep	resolves	4,313 @ $n_{\text{iter}}=24$; 295 @ $n_{\text{iter}}=28$	structural wall

Table 2: Case I, the repro instrument’s decision-rule walk on the real cell ($K=84$): each step ran in its pre-registered precedence order, and the terminal verdict (tolerance miscalibration) was reachable only after every instrument-bug branch failed to fire. All 4,608 flagged episodes resolve by $n_{\text{iter}} = 28$; none remained at larger budgets. Evidence row I3.

tap site	contender		ablation	
	rf@0.9	gap	rf@0.9	gap
(i) S_1 , shallow query (deployed)	0.0	+0.060	0.0	+0.005
(ii) S_0 , bind pathway	0.0	+0.006	0.0	+0.002
(iii) S_1 , true query pathway	0.0	+0.063	0.0	+0.005
(iv) pre-LM-head hidden	0.674	+0.800	0.0	+0.006

Table 3: Case II, ridge-probe recovery by tap site; “gap” is each fit’s cosine-mean margin over its own shuffled-tap control. No state-level linear tap recovers anything in either arm, including (ii) on the causally load-bearing state; the post-nonlinearity tap (iv) recovers strongly and only in the arm that performs the task. Evidence row W4.

group	ceiling (X)	far (X)	bar	clears	baseline far (X)	separates
S3	0.920	0.900	0.828	yes	0.150	no
S4	1.000	1.000	0.900	yes	0.000	yes
A5	0.490	0.390	0.441	no	0.000	no
S5 (decisive)	0.817	0.483	0.735	no	0.000	no
A6 (decisive)	0.190	0.000	0.171	no	0.000	no

Table 4: Case III, the re-metric: the pre-registered endpoint computed under the crosscheck lens (X) per group, by the unmodified verdict machinery on shadow cells; “far” is depth 64 and “bar” is 90% of each group’s own ceiling. The endpoint remains FALSIFY (neither decisive group separates), but the failure geometry is now informative: A6 never converges at its decisive configuration under either lens, and S5 is held under its bar by one non-generalizing seed. Evidence row X5.

converged control checkpoint	real	shuffled	falsifier (≥ 0.5) fires
A6, $n_h=4$, seed 0	1.000 (bit-identical)	0.000	no
S5, $n_h=4$, seed 0	0.800 (bit-identical)	0.000	no
S4, $n_h=2$, seed 2	1.000 (bit-identical)	0.050	no

Table 5: Case III, the tiebreak’s own pre-registered falsifier: on three converged checkpoints, the crosscheck re-scored with targets shuffled across the item pool reads far below the 0.5 void threshold, while the unshuffled reads reproduce the committed values exactly (“bit-identical”). Had any shuffled read cleared 0.5, the tiebreak was pre-registered to be wrong and the adjudication void. Evidence row X4.